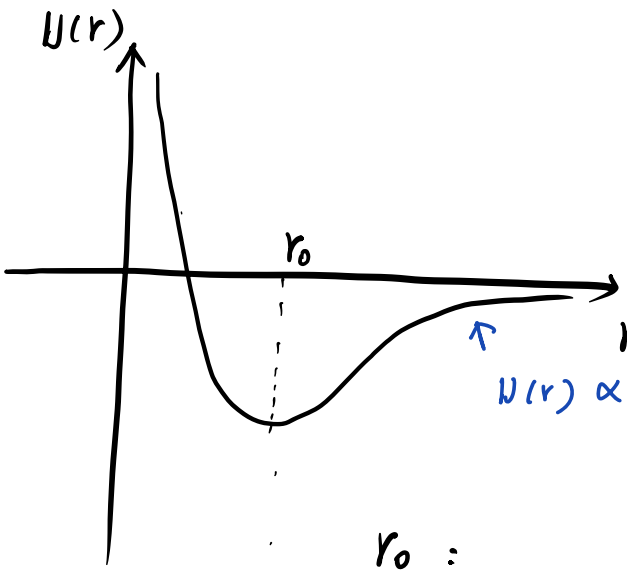
分子力势能曲线



$U(r) \propto 1/r^6$ (if $r \gg r_0$)

r_0 : $\frac{dU(r)}{dr} = 0$

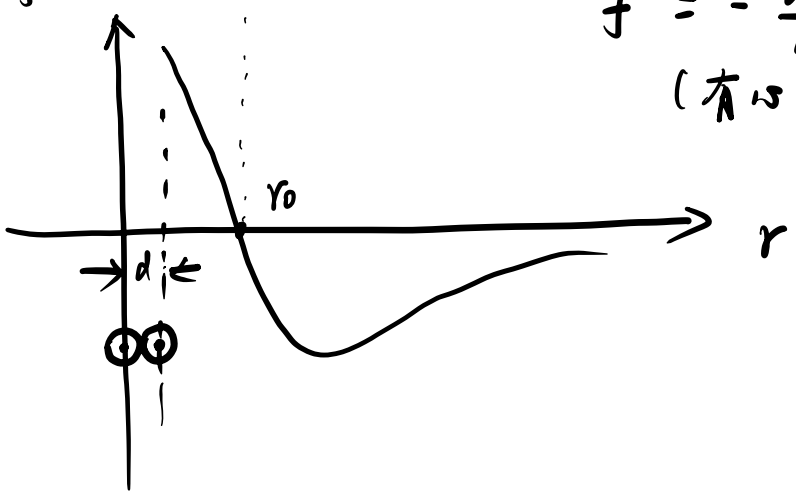
平衡位置

分子力

$\vec{f} = -\nabla U$

$f = -\frac{dU}{dr}$

(有斥力)



elastic force :

$E_p = \frac{1}{2} k x^2$

$F = -\nabla E_p = -kx$

Gravity :

$E_p = mgz$

$F = -E_p = -mg$ ↓ more general

$E_p = -\frac{GmM}{r}$

$\vec{F} = -\nabla E_p = -\frac{GmM}{r^2}\hat{r}$

electrical :

$\vec{F} = q\vec{E}$, $U = q\phi$, voltage

$\vec{E} = -\nabla U$ (U : potential)

已：$\vec{F} = \vec{i}\,x^2y^3 + \vec{j}\,xy^2$

是一个保守力吗？

if $\vec{F}$ is a conservative force

$$F_x = -\frac{\partial U}{\partial x}, \quad F_y = -\frac{\partial U}{\partial y}$$

观察 $\vec{i}$ 分量

$$U \sim x^3y^3$$

观察 $\vec{j}$ 分量

$$U \sim xy^3$$

no such U to satisfy the two components

$$\frac{\partial U}{\partial y\,\partial x} = -\frac{\partial F_x}{\partial y}$$

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$$\frac{\partial U}{\partial x \partial y} = -\frac{F_y}{\partial x} \quad \Rightarrow$$

$$\frac{F_x}{\partial y} = \frac{F_y}{\partial x} \quad (\text{判据})$$

势能函数仅依赖位置

dE_p ：在数学上意味着

E_p 一个全微分

这样的积分函数将与路径无关

积分结果仅与始终点，即始末相关

$$\vec{F} \cdot d\vec{r} = -\nabla E_p \cdot d\vec{r}$$

$$= -\left(\frac{\partial E_p}{\partial x}, \frac{\partial E_p}{\partial y}, \frac{\partial E_p}{\partial z}\right) \cdot (dx, dy, dz)$$

$$= -dE_p$$

泰勒微分 again ^^

Linear approximation

$$F(y) = F(y_0) + F'(y_0)(y-y_0)$$

Taylor expansion:

$$F(y) = F(y_0) + F'(y_0)(y-y_0) + \frac{1}{2}F''(y_0)(y-y_0)^2 + \cdots + \frac{1}{n!}F^n(y_0)(y-y_0)^n + \cdots$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} F^{(n)}(y_0)(y-y_0)^n$$

y 在 y_0 邻域，$\Delta y = y - y_0$ (smallness)

$(\Delta y)^n$：高阶(n)小量

$$\nabla \equiv \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$$

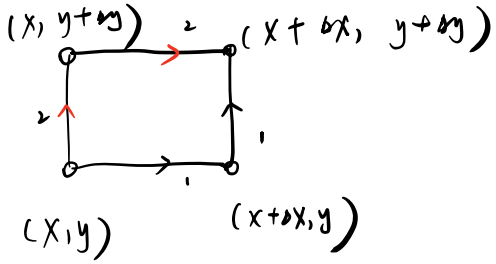
$$\frac{\partial f(x,y,z)}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y, z) - f(x,y,z)}{\Delta x}$$

partial derivative (here, y, z keep invariant)

gradient (∇)：上升最快的方向 （势函数）

$\vec{F} = \ominus \nabla E_p$

$\vec{F}$：其方向是势的下降最快的方向



路径1：$f(x+dx, y+dy) - f(x,y)$

$= f(x+dx, y+dy) - f(x+dx, y)$

$+ f(x+dx, y) - f(x,y)$

$= \left.\frac{\partial f}{\partial y}\right|_{x+dx,y} dy + \left.\frac{\partial f}{\partial x}\right|_{x,y} dx$

① ②

路径2：$f(x+dx, y+dy) - f(x,y)$

$= f(x+dx, y+dy) - f(x, y+dy)$

$$+ f(x, y+\Delta y) - f(x, y)$$

$$= \left.\frac{\partial f}{\partial x}\right|_{x, y+\Delta y} \Delta x + \left.\frac{\partial f}{\partial y}\right|_{x, y} \Delta y$$

③ ④

$$① + ② - ③ - ④ = 0$$

$$① - ④ = \frac{\partial}{\partial x}\frac{\partial f}{\partial y} \Delta x \Delta y$$

$$② - ③ = -\frac{\partial}{\partial y}\frac{\partial f}{\partial x} \Delta y \Delta x$$

$$\Downarrow$$

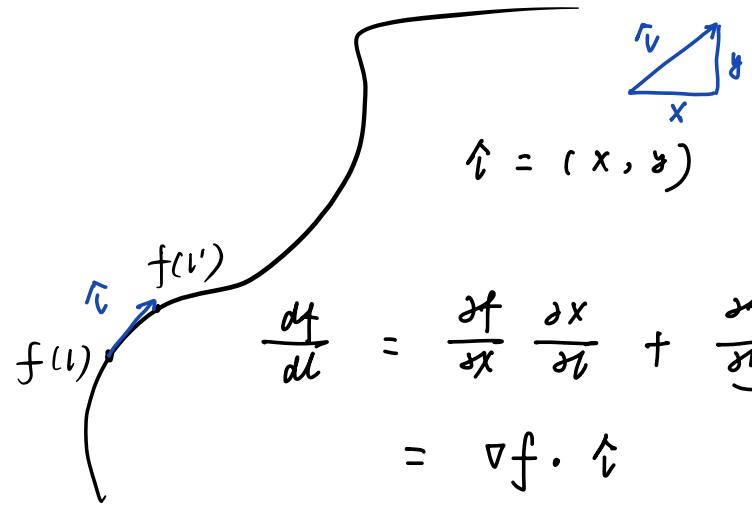
$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

$$\frac{\partial f}{\partial x} = -F_x, \quad \frac{\partial f}{\partial y} = -F_y$$

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}$$

方向导数：(directional derivative)

$\frac{df}{d\vec{l}}$



$$\frac{df}{dl} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial l} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial l}$$

$$= \nabla f \cdot \hat{l}$$

① $= \lim_{l \to l'} \frac{f(l') - f(l)}{\Delta l}$ $\hat{l}$

$f(l') - f(l) = \Delta f(l)$

$= \nabla f(l) \cdot \underline{d\vec{r}}$ $(l \to l')$

② $= (\nabla f \cdot \hat{l})\,\hat{l}$ $\qquad \frac{\nabla f(l) \cdot d\vec{r}}{\Delta l} = \nabla f(l) \cdot \hat{l}$

E.g. $\quad E_k = \frac{1}{2} m v^2 = \frac{1}{2} m \vec{v} \cdot \vec{v}$

计算 $\frac{\partial E_k}{\partial \vec{v}} = \frac{1}{2} m \left(\frac{\partial v^2}{\partial \vec{v}}\right)$

$= \frac{1}{2} m (\nabla v^2 \cdot \hat{v})\,\hat{v}$ (用等号"="②)

$$= \frac{1}{2} m \left[(2v_x \hat{i} + 2v_y \hat{j} + 2v_z \hat{k}) \cdot \hat{v} \right] \hat{v}$$

$$= m \left[\vec{v} \cdot \hat{v} \right] \hat{v}$$

$$= m v \hat{v} = m\vec{v}$$

或者 $E_k = \frac{1}{2} m v^2$

$$\frac{\partial E_k}{\partial \vec{v}} = \frac{\partial E_k}{\partial v} \hat{v} \quad (\text{因为} = v \text{①})$$

根据定义 $= m v \hat{v} = m\vec{v}$

$$\int \vec{F} \cdot d\vec{r} = \int m\vec{a} \cdot d\vec{r}$$

$$= \int m \frac{d\vec{v}}{dt} \cdot d\vec{r}$$

$$= \int m\vec{v} \cdot d\vec{v}$$

$$= \int d\left(\frac{1}{2} m v^2\right) = \Delta E_k$$